

# GeCCo Paper Study Summary

Experiments 1–4 + Logistic Function Explained with Examples

# 1. Big Picture: What Is GeCCo Doing?

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Traditional cognitive modeling:

researcher proposes theory → writes model → fits  
human data → compares models

GeCCo:

LLM reads task + data + constraints → generates  
model code → model is fitted → feedback improves  
next model

The goal is not only prediction.

The goal is to find simple, interpretable explanations of  
behavior.

## Key idea

LLM-generated models are tested against human behavioral  
data, not accepted automatically.

## Why it matters

It broadens the search space of possible cognitive theories.

## 2. Experiment 1: Decision Making

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### Task

Participants choose between two options.

Each option has binary features: 0 / 1.

Each feature has a validity: how reliable the feature is.

Question: how do people combine multiple cues?

### Result

Baseline model: pWADD.

Best LLM models outperformed the literature baseline.

The LLM model used feature validities plus a discount factor.

Main lesson: simple models can explain human choices well.

### 3. Experiment 1 Example: Weighted Choice

#### Option score

$$\text{score}(A) = \sum \text{feature} \times \text{validity}$$

If Option A has strong valid features, its score becomes higher.

#### Choice probability

$$P(\text{choose } A) = \text{logistic}(\text{score}A - \text{score}B)$$

The score difference is converted into a probability between 0 and 1.

Example:  $\text{score}(A) = 3$ ,  $\text{score}(B) = 1$

Difference = 2  $\rightarrow$   $\text{logistic}(2) \approx 0.88$

Interpretation: the model predicts an 88% probability of choosing A.

## 4. Experiment 2: Learning

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### Task

Participants repeatedly choose between options.  
They receive feedback after each choice.  
They learn which option is more rewarding.  
This is similar to a bandit learning task.

### Result

Baseline: Rescorla–Wagner model variant.  
The LLM model learned from chosen outcomes.  
It also used forgone / counterfactual outcomes.  
Main lesson: people learn from both what happened  
and what could have happened.

## 5. Experiment 2 Example: Prediction Error

### Learning rule

$$\text{new value} = \text{old value} + \alpha \times \text{prediction error}$$

**error**

$\alpha$  is the learning rate. It controls how strongly the model updates after feedback.

### Prediction error

$$\text{PE} = \text{reward} - \text{expected value}$$

If the result is better than expected, PE is positive; if worse, PE is negative.

Example: expected value of A = 0.4, reward = 1,  $\alpha$  = 0.5

$$\text{PE} = 1 - 0.4 = 0.6$$

$$\text{new value} = 0.4 + 0.5 \times 0.6 = 0.7$$

## 6. Experiment 3: Planning

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### Task

Classic two-step task.

Participants make a first-stage choice.

The choice leads to a second-stage state.

Transitions can be common or rare.

### Result

Baseline: hybrid model.

Classic theory: model-free + model-based control.

LLM model captured transition-dependent learning.

Main lesson: planning may be explained through learned transition structure.

## 7. Experiment 3 Example: Common vs. Rare Transition

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Suppose action A usually leads to state X.

If  $A \rightarrow X$  happens, it is a common transition.

If  $A \rightarrow Y$  happens, it is a rare transition.

A planning model asks:

Did the reward come from the action itself, or from an unusual transition?

### Model-free intuition

If action A was rewarded, repeat A.

### Model-based intuition

Use the transition structure to decide whether A is actually good.

### LLM insight

The generated model tracked common and rare transitions separately.



## 8. Experiment 4: Working Memory

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### Task

Participants learn stimulus-action associations.

Low load: fewer associations.

High load: more associations.

The task separates reinforcement learning and working memory.

### Result

Baseline: RL-WM model.

Low load: faster, more memory-like learning.

High load: slower, more RL-like learning.

Main lesson: LLMs can capture load-dependent learning.

## 9. Experiment 4 Example: Cognitive Load

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Low load example:

Learn 3 stimulus-action mappings.

Working memory can hold them, so learning is fast.

High load example:

Learn 6 stimulus-action mappings.

Working memory is limited, so behavior relies more on slow reward learning.

### Interpretation

The LLM model captured the behavioral pattern without necessarily copying the full RL-WM architecture.

### Teaching point

Similar behavior can sometimes be explained by different computational mechanisms.

## 10. Logistic Function: Why Do We Need It?

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Cognitive models often compute a score.

But a score can be any real number:

-10, -2, 0, 3, 100 ...

A probability must be between 0 and 1.

So we need a function that turns an unbounded score into a valid probability.

**Logistic function**

$$P = 1 / (1 + e^{(-x)})$$

Input x can be any real number. Output P is always between 0 and 1.

# 11. Logistic Function: Three Key Cases

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## **x is very negative**

Example:  $x = -10$

$P \approx 0$

Interpretation: choosing A is very unlikely.

## **x = 0**

$P = 0.5$

Interpretation: both options are equally likely.

## **x is very positive**

Example:  $x = 10$

$P \approx 1$

Interpretation: choosing A is very likely.

This is why logistic is often used for binary choices: choose A vs. choose B.

## 12. Logistic Example: From Value Difference to Choice

### Choice model

$$P(A) = \text{logistic}(\text{valueA} - \text{valueB})$$

The model first compares the two option values.

### Example 1:

$$\text{valueA} = 3, \text{valueB} = 1$$

$$x = 3 - 1 = 2$$

$$P(A) = \text{logistic}(2) \approx 0.88$$

### Example 2:

$$\text{valueA} = 1, \text{valueB} = 3$$

$$x = -2$$

$$P(A) = \text{logistic}(-2) \approx 0.12$$

# 13. Logistic vs. Softmax

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## Logistic

Used mainly for two choices.

Input: one score difference  $x$ .

Output: probability of one option.

Example:

$P(A) = \text{logistic}(\text{valueA} - \text{valueB})$

## Softmax

Used for two or more choices.

Input: a set of scores.

Output: a probability distribution whose probabilities sum to 1.

Example:

$P(A), P(B), P(C)$

# 14. Softmax Example: Three Options

## Softmax

$$P(i) = \exp(\text{value\_i}) / \sum \exp(\text{value\_j})$$

Softmax turns multiple option values into probabilities that add up to 1.

Example values:

$$A = 3, B = 1, C = 0$$

Approximate probabilities:

$$P(A) \approx 0.84$$

$$P(B) \approx 0.11$$

$$P(C) \approx 0.04$$

Interpretation: A is most likely, but the model still allows uncertainty.

## 15. Why Logistic Matters for This Paper

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The paper studies human choices and learning behavior.

Many cognitive models compute internal values, scores, or preferences.

But observed behavior is probabilistic: people do not always choose the same option.

Logistic and softmax functions connect hidden model values to observable choice probabilities.

This makes the model testable: predicted probabilities can be compared with real human choices.



# 16. Summary of Experiments 1–4

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## Experiment 1 Decision making

How people combine binary features and validities.

LLM found a simple weighted-choice model.

## Experiment 2 Learning

How people update choices from feedback.

LLM separated chosen and counterfactual outcomes.

## Experiment 3 Planning

How people use transition structure.

LLM captured common vs. rare transition learning.

## Experiment 4 Working memory

How cognitive load affects learning.

LLM captured fast low-load and slow high-load learning.

## 17. Take-home Message

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GeCCo uses LLMs to generate computational cognitive models.

The models are not accepted by intuition; they are tested with behavioral data.

Experiments 1–4 show that LLM-generated models can match or outperform handcrafted baselines.

Logistic and softmax functions are important because they turn internal scores into choice probabilities.

The key learning point: cognitive models connect theory, computation, and human behavior.